

Let Us Learn HOW TO PROGRAM 8085

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Study of microprocessors is an integral part of the engineering programme, when we are taking up a course on electronics or any related discipline. And 8085 is probably the first microprocessor that we get in touch with as a part of the curriculum. Students are usually just given some basic hands-on training using standard boards with a keypad.

Whenever we are dealing with hardware, there is always a chance for it to fail. Moreover, there are some practical concerns while learning microprocessors using boards. If you would like to simultaneously determine the status of all the registers at a time, or analyse the working in varying speed of simulations, a trainer kit may have limitations.

For electronics enthusiasts willing to learn programming of microprocessors, but limited by the availability of trainer kits, there are a handful of simulators available online as open source. In this article, let us have a look at one such 8085 simulator.

The 8085 programming model

8085 processor has a set of seven 8-bit registers including the accumulator and six others, namely, B, C, D, E, H and L. Depending upon applications, the registers other than the accumulator can be used either as independent byte-registers or as 16-bit register pairs.

A 16-bit special-purpose register called program counter is available in the microprocessor. It stores the address of the next instruction to be fetched. A 16-bit stack pointer stores



Screenshot of 8085 simulator

Software at a glance

Name	: 8085 Simulator version 2.0
Licence	: GNU GPLv2
Pre-requisites	: Needs Java 6 update 16; available from javadt.sun.com/webapps/download/AutoDL?BundleId=33889

the address of the last byte entered into the stack.

Program Status Word comprising the accumulator and status registers is modified as per given instructions. An accumulator stores the result of an arithmetic and logical operation, and the result affects the content of various status registers.

What the simulator consists of

The simulator is designed to be very user-friendly and simulations are to be at par with actual hardware

simulations. Let us have a quick look at the interesting features offered by the tool.

Assembler editor. This allows the user to input numerical data in binary, decimal and hexadecimal formats. The programmer can insert comments, label instructions and check errors using the editor. There are provisions for auto-correct, auto-intend and syntax highlighting. Moreover, users are allowed to run programs written in other simulators in this editor.